Exponential amplification in the dS_3 crossing kernel: full-factor tail bounds and an obstruction to direct summation

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Abstract

An accurate approximation to one crossing factor need not give an accurate physical matrix element. We construct exact rational polynomial certificates for the tails of all four Omega factors in the known scalar LPRS formula, including the digamma series at a removable collision. We propagate the bounds through the two shadow terms and derive an elementary expression for their coefficient moduli. One coefficient grows as $\exp(2\pi M)$ in the specified heavy-observer limit. Direct partial sums can therefore fail by many orders of magnitude even when their arithmetic is stable and their individual absolute tails are small. This identifies an analytic obstacle; it does not establish a heavy-mass distributional limit or a de Sitter holographic dual.

1 Scope and a general series certificate

We retain radius one, $G_N = 0$, proper $SO^+(3,1)$, BD scalar observers with a fixed positive mass-label lower bound, KG-normalized matter, and the previously defined half-period regulated local creation blocks. The object is the scalar-to-scalar element U_{00} of the full Racah transform. The scalar subblock is not asserted to be unitary. External labels are (a, q, c, p) and internal labels (b, d), all positive. The chiral weights obey $h_i = (1 + i\nu_i)/2$ with $\bar{h}_i = h_i$, not complex conjugation.

Every ordinary hypergeometric series needed in LPRS (3.38) has terms

$$t_n = \frac{\prod_{j=1}^{4}(\frac{1}{2} + i\alpha_j)_n}{n! \prod_{k=1}^{3}(1 + i\beta_k)_n}, \qquad \alpha_j, \beta_k \in \mathbb{R}, \quad t_0 = 1.$$

The definition is standard [2]. Set

$$A(x) = \prod_j [(x + \tfrac{1}{2})^2 + \alpha_j^2], \quad B(x) = (x+1)^2 \prod_k [(x+1)^2 + \beta_k^2],$$

$$P(x) = B(x)(x+1)^3 - A(x)(x+2)^3.$$

The degree-eleven terms cancel; P has degree ten and leading coefficient one, independent of the parameters. Choose an integer N_0 for which all coefficients of $P(N_0 + y)$ are nonnegative. This implies

$$\left|\frac{t_{n+1}}{t_n}\right| \le \left(\frac{n+1}{n+2}\right)^{3/2}, \qquad n \ge N_0. \tag{1}$$

Such a shift exists for every fixed parameter set, since the derivatives of a positive-leading polynomial are eventually positive. For rational imaginary parts we check the coefficients with exact rational arithmetic. This is a sufficient certificate, not a minimal cutoff or a bound uniform in growing masses.

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2 Tails at and away from a collision

For $N \geq N_0$, (1) implies $|t_n| \leq |t_N|((N+1)/(n+1))^{3/2}$ for $n \geq N$. The integral test gives the exact-arithmetic tail bound

$$\left|\sum_{n=N}^{\infty} t_n\right| \leq |t_N|(2N+3). \tag{2}$$

Unlike an asymptotic zeta acceleration, no discarded inverse-power remainder is involved. The price can be a very loose bound.

At an exact collision let H be either h or $1-h$, $h_0' = H + h_1 + h_3 - 1$, and

$$(A_1, A_2, A_3, A_4) = (H + h_1 + h_2 - 1, H + h_3 + h_4 - 1, H + h_3 - h_4, H + h_1 - h_2),$$

$$(B_1, B_2, B_3, B_4) = (2h_0', 2H, 1, 1).$$

The preceding merged-series derivation gives

$$\Omega_0 = \sum_{n=0}^{\infty} t_n q_n, \qquad q_n = \sum_{j=1}^{4} [\psi(n + B_j) - \psi(n + A_j)],$$

where the t_n denominator is $(2h_0')_n (2H)_n (n!)^2$. All A_j have real part $1/2$ and all B_j real part one. Using the trigamma series [3] along the straight segment between the two arguments,

$$|q_n| \leq L\left[\frac{1}{n+1/2} + \frac{1}{(n+1/2)^2}\right], \qquad L = \sum_j |B_j - A_j|.$$

Define $C_N = L(1 + (N+1/2)^{-1})(N+1)/(N+1/2)$. A second integral test yields

$$\left|\sum_{n=N}^{\infty} t_n q_n\right| \leq |t_N| C_N \left[\frac{1}{N+1} + \frac{2}{3}\right]. \tag{3}$$

The implementation advances q_n using the digamma recurrence, retaining all complex phases. Away from collision the two ordinary series receive their exact Gamma coefficients, and their two bounds are added in modulus. That bound need not remain useful as the collision is approached.

Write the complete physical element as

$$U_{00} = C_+ \Omega(h, h')\Omega(1-h, h') + C_- \Omega(h, 1-h')\Omega(1-h, 1-h').$$

For approximate factors x_σ, y_σ with tail bounds $e_{x,\sigma}, e_{y,\sigma}$, the propagated bound is

$$|U_{00} - \widehat{U}_{00}| \leq \sum_{\sigma=\pm} |C_\sigma|(|x_\sigma| e_{y,\sigma} + |y_\sigma| e_{x,\sigma} + e_{x,\sigma} e_{y,\sigma}). \tag{4}$$

Equations (2)–(4) concern exact finite sums and exact Gamma coefficients. Evaluating their right-hand sides with mpmath is not an interval certificate for rounding or special functions.

3 The exponentially amplified shadow coefficient

The coefficients include the physical phase normalization of our earlier crossing dictionary, whose modulus is bd/π^4. LPRS (3.35)–(3.38) [1] then imply an elementary modulus formula. Define

$$x_1 = \frac{b+a-q}{2}, \quad x_2 = \frac{b-a+q}{2}, \quad x_3 = \frac{-b+p-c}{2}, \quad x_4 = \frac{-b+c-p}{2}.$$

The Gamma reflection identities give

$$|\Gamma(\tfrac{1}{2}+ix)|^2 = \frac{\pi}{\cosh \pi x}, \qquad |\Gamma(1+ib)|^2 = \frac{\pi b}{\sinh \pi b}.$$

The four-Gamma common prefactor has modulus $\pi \sinh(\pi b)/(4b\sqrt{\prod_j \cosh(\pi x_j)})$. Each shadow coefficient has modulus π/d; the remaining Gamma ratios have unit modulus because their half-integer arguments pair with opposite imaginary parts. Finally, $|\sin(-\pi/2+iy)| = \cosh y$. Consequently,

$$|C_\sigma| = \frac{\sinh(\pi b)}{4\pi^2\sqrt{\prod_j \cosh(\pi x_j)}} \cosh\frac{\pi(\sigma d - a - p)}{2} \cosh\frac{\pi(\sigma d - q - c)}{2}. \tag{5}$$

This is an exact modulus identity, not a phase convention or a replacement of the full complex kernel by positive quantities.

Now fix u, v, p, q and take $c = M$, $b = M + u$, $a = M + u + v$, $d = M + v$. Then

$$\begin{aligned}\frac{|C_-|}{|C_+|} = \frac{\exp(2\pi M + \frac{\pi}{2}(u+3v+p+q))}{4\cosh\frac{\pi(u+p)}{2}\cosh\frac{\pi(v-q)}{2}} \\ \times (1+e^{-\pi(d+a+p)})(1+e^{-\pi(d+q+c)}).\end{aligned} \tag{6}$$

Moreover,

$$|C_+| \longrightarrow \frac{e^{\pi(u-v+p+q)/4}}{4\pi^2}\sqrt{\cosh\frac{\pi(u+p)}{2}\cosh\frac{\pi(v-q)}{2}} > 0.$$

Thus $|C_-|$ grows as $e^{2\pi M}$ with a nonzero fixed coefficient. An independent absolute-error estimate for $\Omega(h, 1-h')$ must offset this exponential gain whenever its partner is not comparably small. Polynomial accuracy of a different Omega cannot supply that control.

This does not prove that the physical matrix element diverges. It explains why an exponentially small factor or cancellations must be controlled before taking the heavy-mass limit of this representation.

4 Numerical evidence, limitations, and next calculation

We use $p = q = 0.6$, $u = 0.3$, $v = -0.2$ on the collision surface. All tested polynomial certificates allow $N = 64$; we also sum $N = 128$. The latter sums at 110 and 150 decimal digits agree relatively to better than 10^{-106} for the two larger masses. They are nevertheless poor approximations to the complete physical element:

M	N	observed $\lvert\widehat{U} - U_{\mathrm{ref}}\rvert$	full tail bound
2	128	$2.862\,10^{-3}$	$3.589\,10^{-1}$
8	128	$1.064\,10^{7}$	$3.391\,10^{8}$
32	128	$4.982\,10^{43}$	$7.527\,10^{44}$

The references are the previously saved complete-kernel calculations. The $M = 2$ reference retains its known prototype bias; no reference is being treated as an exact interval enclosure. These observed errors are far larger than that bias. All nine runs, including the failed approximations, are preserved.

At $M = 32$, the modulus ratio in (6) is approximately $10^{86.7155}$. The computed tail of the first Omega in the minus-shadow term is only about $4.73\,10^{-42}$, but its amplification is enormous. Holding the other computed factor fixed, an absolute 10^{-6} budget for this product requires an error near $6.28\,10^{-93}$ in that Omega. This is a sensitivity calculation at the stored finite sum, not a certificate that the exact partner or complete product has that error.

Independent Fraction arithmetic reconstructs every shifted polynomial; Decimal arithmetic replays each first-omitted modulus, ordinary or weighted tail, full-product propagation, and complex reference residual. All 407 checks pass. Passing these checks verifies the bound machinery and the retained failures, not the accuracy of these direct sums. The exact coefficient identity is independently checked against the Gamma expression at 150 digits. Versions and source/input hashes accompany the saved results.

The preceding collision note's separate 100-digit hypergeometric parameter-derivative validation remains running at this checkpoint. It is neither required for the rational certificates here nor counted as a passed check. Both stages remain unfrozen pending final project integration.

The next analytic problem is the exponentially small $\Omega(h, 1-h')$ on this physical slice, with its phase and multiplying coefficient kept together. Candidates are an Euler-contour deformation or a complete-kernel integral in which the large coefficient has already been combined. A fixed-label tail certificate alone cannot justify an exchange with the physical mass integrals. The full crossed heavy-observer limit, Bose free limit, full Fock domain, finite-G_N dynamics, and microscopic holographic dictionary remain open.

References

[1] J. Liu, E. Perlmutter, V. Rosenhaus and D. Simmons-Duffin, *d-dimensional SYK, AdS Loops, and* $6j$ *Symbols*, arXiv:1808.00612v3, Eqs. (3.35)–(3.38).

[2] NIST DLMF, Sec. 16.2, generalized hypergeometric series.

[3] NIST DLMF, Sec. 5.15, trigamma series; Sec. 5.4, Gamma modulus identities. Consulted 31 October 2026.